FBISE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

BIOLOGY

Past Paper 2023 • Paper Code: 1HA-I 2310 — B(3104) / D(7101)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

Note: BIOLOGY HSSC-I. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 3104. Fill the relevant bubble against each question according to curriculum:

Q1. A person is given vaccine of COVID. This is an example of:

(A) Artificial Active immunity
(B) Natural Active immunity
(C) Artificial Passive immunity
(D) Natural Passive immunity

Q2. Which structure is the genetic material of the virus? (Note: a diagram of a bacteriophage is given with labels I = head, II = inside of head, III = tail sheath, IV = tail fibres.)

(A) II
(B) III
(C) IV
(D) I

Q3. Advantages to life on land for early amphibians included:

(A) Greater temperature variation
(B) More food and no predation
(C) Less O2 in the air than water
(D) More food and more predation

Q4. A group of ribosomes attached to mRNA is known as:

(A) Glyoxisomes
(B) Polysomes
(C) Lysosomes
(D) Peroxisomes

Q5. Which of the following pair is mismatched?

(A) Fat – lipids
(B) Amino acid – protein
(C) Starch – nucleic acid
(D) Sugar – carbohydrate

Q6. What can be found in both DNA and messenger RNA?

(A) Sugar – phosphate chain
(B) Ribose
(C) Thymine
(D) Double helix structure

Q7. The compound that enters Kreb's cycle from glycolysis is:

(A) Oxaloacetate
(B) Pyruvic acid
(C) Acetyl-CoA
(D) Citric acid

Q8. Mesosomes are infolding of the cell membrane and are involved in:

(A) Protein synthesis
(B) RNA synthesis
(C) Metabolism
(D) DNA replication

Q9. Chitin is found in the cell wall of fungi, which is also found in exoskeleton of:

(A) Echinoderms
(B) Molluscs
(C) Chordates
(D) Arthropods

Q10. The process illustrated in the figure is called: (Note: a diagram shows a spore producing hyphae.)

(A) Fertilization
(B) Germination
(C) Host invasion
(D) Photosynthesis

Q11. In spermatophyte, seed is formed from:

(A) Ovary
(B) Anther
(C) Embryo sac
(D) Ovule

Q12. Which of the following pair of words are correctly matched?

(A) Scraping algae – siphon
(B) Predation – tentacle
(C) Nephridia – gizzard
(D) Filter feeding – radula

Q13. Apical dominance is caused by:

(A) Cytokinin
(B) Ethene
(C) Auxins
(D) Gibberellins

Q14. A diagram of villus is given. Which labelled part is a vessel of lymphatic system? (Note: labels I, II, III, IV are marked on a villus diagram.)

(A) II
(B) III
(C) IV
(D) I

Q15. Which heart chamber pumps blood to the lungs? (Note: a heart diagram with chambers labelled 1, 2, 3, 4 is given.)

(A) 2
(B) 3
(C) 4
(D) 1

Q16. In which of the following blood vessels the blood pressure is highest?

(A) Arteries (B) Capillaries

(C) Veins (D) Aorta

Q17. Which of the following cells are destroyed in an AIDS infection?

(A) Helper T cells (B) B cells

(C) Memory cells (D) Macrophages

> Note: BIOLOGY HSSC–I. Time allowed: 2:35 Hours. Total Marks Sections B and C: 68. NOTE: Answer any fourteen parts from Section 'B' and attempt any two questions from Section 'C' on the separately provided answer book. Write your answers neatly and legibly.

Q. SECTION – B (Marks 42)

Q. 2 Attempt any FOURTEEN parts from the following. All parts carry equal marks. (14 x 3 = 42)

(i) List any three organelles bounded by a single membrane and any three organelles bounded by double membrane.

(ii) State property of water that allows each of the following to take place. Briefly explain its importance. a. The cooling of skin during sweating b. The survival of fish in ice-covered lake

(iii) The graph shows the effect of temperature on the rate of reaction of an enzyme. (Note: a graph of Rate of Reaction vs Temperature (°C) is given, rising to a peak marked X at about 35–37°C then falling; region A is the rising part, region B the falling part.) a. What is indicated by X? b. What is happening in region A? c. What is happening in region B?

(iv) Briefly explain what will happen if dead bodies are not decomposed by bacteria?

(v) What will happen if plants are exposed to green light? Explain briefly.

(vi) How penicillin affects bacterial cell? Describe briefly.

(vii) Complete the concept map with the following terms: Amoeba, Protozoan, Flagellates, Sporozoans, Ciliates, Paramecium. (Note: a concept map reads 'Animal like protists are called (1)', 'Which include (2) That use Pseudopodia, (3) That use cilia, (4) That use flagella, (5) That produce spores', with (6) below (3).)

(viii) List six features of bryophytes which made them to adapt on land.

(ix) Why many insects are pests to humans when they are larvae but are beneficial when they are adult? Explain briefly.

(x) Complete concept map: Prototherian, Metatheria, Eutherian, Placenta, Kangaroo, Duckbill platypus. (Note: a concept map reads 'Three groups of class Mammalia include (1) That have a pouch → Example (4); (2) That lay eggs → Example (5); (3) That have a/an (6)'.)

(xi) Differentiate between. a. Collenchyma and sclerenchyma b. Photoperiodism and Phototropism c. Apoplast pathway and symplast pathway.

(xii) Many animals including humans eat seeds such as beans, peas, peanuts and almonds. Why are seeds a good food source?

(xiii) If stomach fails to produce HCl, how would this affect digestion? If left untreated how could this affect the body as whole?

(xiv) Describe three functions of digestive system. Write down sequence of organs according to the order in which food passes through them.

(xv) How would chronic diarrhoea affect homeostasis of the body?

(xvi) What type of chemical reaction would be involved in the formation of glucose from starch or glycogen? Explain briefly.

(xvii) Complete the concept map with the given terms: Aorta, Left atrium, Left ventricle, Right atrium, Right ventricle, Lungs. (Note: a concept map reads 'Blood moves from vena cava ↓ (1) ↓ (2) ↓ Pulmonary artery ↓ (3) → Pulmonary veins → (4) of heart ↑ (5) ↑ (6) ↑ To the body'.)

(xviii) A new mother had chicken pox as a child. Why does not her new born child get the disease, even after being exposed to its virus?

(xix) How photosynthetic pigments are arranged in photosystem? Differentiate between the two types of photosystem.

(xx) Differentiate between viruses and viroid. Also give examples

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. What is plasma membrane? Explain in detail fluid mosaic model of plasma membrane. b. How does body give inflammatory response when tissues are damaged?

Q. 4 a. i. How does a bacterium reproduce sexually? Briefly explain. ii. Photosynthesis and respiration are said to be complementary. How are these chemical reactions connected? b. Describe uses of fungi in genetic engineering and antibiotic production.

Q. 5 a. What does ECG stand for? Explain different phases with the help of labelled diagram. b. Compare the osmotic adjustments in different groups of plants in different environments.

> Note: — 1HA-I 2310 —. Page 1 of 2 (Biology). Page 2 of 2 (Biology).

VERSION D

> Note: BIOLOGY HSSC-I. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 7101. Fill the relevant bubble against each question according to curriculum:

Q1. Fluidity of cell membrane is due to:

(A) Lipids
(B) Proteins
(C) Carbohydrates
(D) Vitamins

Q2. Which term describes both collagen and hemoglobin?

(A) Enzyme
(B) Fibrous Protein
(C) Globular Protein
(D) Macromolecule

Q3. The diagram below shows an enzyme and an inhibitor. Which of the following describe the inhibitor? (Note: a diagram shows an inhibitor that does not fit into the enzyme's active site.)

(A) Competitive, irreversible
(B) Non competitive reversible
(C) Non competitive, irreversible
(D) Competitive, reversible

Q4. During dark reactions of photosynthesis:

(A) Water is split off
(B) Chlorophyll is activated
(C) CO_2 is reduced to organic compound
(D) Glucose is broken down

Q5. Which of the following cells are responsible for antibody production?

(A) Macrophages
(B) T cells
(C) B cells
(D) Plasma cells

Q6. Which of the following may build coral reefs along with coral animals?

(A) Myxomycota
(B) Red algae
(C) Brown algae
(D) Green algae

Q7. In given diagram of fungi structure 1 and structure 2 represent: (Note: a diagram labels a spore as 1 and hyphae as 2.)

(A) Fungal capsule and mycelium
(B) Hyphae and lichen
(C) Spore and hyphae
(D) Lichen and mycelium

Q8. Which part of flower develops into fruit?

(A) Ovary wall
(B) Ovule
(C) Anther
(D) Stigma

Q9. A butterfly excretes its wastes through:

(A) Spinnerter
(B) Malphighian tubules
(C) Spiracles
(D) Tracheal tubes

Q10. The cambium and meristems are examples of ____ tissues.

(A) Support
(B) Protective
(C) Growth
(D) Transport

Q11. The given diagram shows the details of gastric gland in stomach wall. Which labelled part is concerned with secretion of HCl? (Note: a gastric gland diagram with parts labelled 1, 2, 3, 4 is given.)

(A) 1
(B) 2
(C) 3
(D) 4

Q12. The given diagram shows chambers of heart. Chamber labelled as 1 receives blood from which blood vessel? (Note: a heart diagram with chambers labelled 1, 2, 3, 4 is given.)

(A) Aorta
(B) Vena Cava
(C) Pulmonary arteries
(D) Pulmonary veins

Q13. The lymphatic system performs which of the following functions?

(A) Transport tissue fluid back into blood stream
(B) Carries O2 and CO2 to and from body cell
(C) Control metabolism
(D) Exchange gases between blood and air

Q14. When a pathogen enters the body, usually which cells are the first to attack it?

(A) Macrophages
(B) T cells
(C) B cells
(D) Plasma cells

Q15. Bacterial membrane differs from Eukaryotic membrane in lacking:

(A) Proteins
(B) Lipids
(C) Polysaccharide
(D) Cholesterol

Q16. Which of the following bacteria are correctly matched for their energy requirements? (Note: a table is given — I. Light – Obligate Saprophytes; II. Hydrogen – Archaebacteria; III. Oxygen – Obligate anaerobes; IV. Oxygen – Obligate aerobes.)

(A) I
(B) II
(C) III
(D) IV

Q17. Which of the following characters do all vertebrates have in common at some point in their development? (Note: a table is given — I. Bilateral symmetry / Exoskeleton / Lungs; II. Back bone / Postanal tail / Lungs; III. Pharyngeal pouches / Notochord / Backbone; IV. Closed circulatory system / Segmented bodies / External fertilization.)

(A) I
(B) II
(C) III
(D) IV

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Q. SECTION – B (Marks 42)

Q. 2 Attempt any FOURTEEN parts from the following. All parts carry equal marks. (14 x 3 = 42)

(i) List any six structural features that are different in Prokaryotic and Eukaryotic cells.

(ii) Complete the table below which shows functional categories of proteins. (Note: a table with columns Category / Example is given; rows: Transport of oxygen, Enzyme, (blank)/ Insulin, Defensive, (blank)/Actin and myosin, Storage.)

(iii) Proteases are used in biological washing powder. How would it remove a blood stain on clothes? Why these washing powders are recommended for use at low washing temperature?

(iv) Compare action spectrum and absorption spectrum of light with reference to photosynthesis.

(v) What is meant by "viruses are obligate intra cellular parasites"?

(vi) Describe the state of HIV virus in a person positive for HIV but not showing any symptom.

(vii) What role do contractile vacuoles play in helping fresh water protozoans?

(viii) Why are fungi placed in a separate kingdom?

(ix) Why bryophytes are called amphibian plants?

(x) When temperature increases and soil becomes drier, what happens to earth worms? How earth worms help to increase soil fertility?

(xi) Which organelle is a site for cellular respiration? Enlist any four of its characteristics.

(xii) Putting a ripened banana in a paper bag with an unriped fruit quickens its ripening. Explain what happens inside the bag.

(xiii) Distinguish between: a. Lysosome – Peroxisomes b. Chloroplast – Chromoplast c. Microtubules – Microfilaments

(xiv) Explain why a person without a gall bladder is unable to digest fat efficiently?

(xv) Complete the concept map with the given terms; Liver, Bile, Small intestine, Stomach, Oesophagus, Gall bladder. (Note: a concept map reads 'Tube likes structure (1) → Passes food to the (2) → For digestion → Which passes food to the (3) ← Which delivers it to the; Gland (4) → Makes (5) → Which is stored in the (6)'.)

(xvi) When strepto coccus bacterium causes severe throat infection, it can get into blood and damage heart valves. How does this affect heart function?

(xvii) The figure below is a diagram of a secretory cell of Pancrease. Name the parts labelled as 1 and 2. Enlist the processes occuring here. (Note: a diagram of a pancreatic secretory cell with parts labelled 1 and 2 is given.)

(xviii) Why is it adaptive for memory cell to remain in the immune system after an invasion by a pathogen?

(xix) Describe the structure and processes involved in innate immunity. How do these differ from acquired immunity?

(xx) In what kinds of environment is it expected to find slime mold? Give four salient features of slime mold.

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. The two strands of DNA are not identical but are complementary. Give reasons. Also explain double helical structure of DNA molecule. b. How do bacteriophages reproduce? Explain lytic and lysogenic cycle in detail.

Q. 4 a. Below is a reaction of Calvin cycle: CO_2 + Ribulose biphosphate → 3 Phospho glycerate. Which enzyme catalyses this reaction? Where does this reaction take place in plants? Explain next two phases of Calvin cycle. b. Explain different modes of sexual reproduction in fungi. Also draw a labelled diagram of life cycle of flowering plant.

Q. 5 a. Explain the conducting system of the heart? Why cardiac muscles are said to be myogenic? b. How the structure of small intestine is adapted for absorption of digested products? Explain.

> Note: — 1HA-I 2310 HA —. Page 1 of 2 (Biology). Page 2 of 2 (Biology).